

MERIDEN MARKHAM, CT, USA

WMO: 725027

Lat: 41.510N

Lon: 72.828W

Elev: 103

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54788

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.6	(c) 9.8	(d) -9.4	(e) 3.3	(f) 9.2	(g) -5.3	(h) 4.2	(i) 15.0	(j) 23.1	(k) 27.3	(l) 20.2	(m) 26.0	(n) 3.9	(o) 320	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 90.5	(d) 73.6	(e) 87.9	(f) 72.4	(g) 85.1	(h) 70.7	(i) 76.5	(j) 86.0	(k) 75.1	(l) 83.2	(m) 73.6	(n) 80.7	(o) 7.8	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	125.1	79.9	72.5	120.8	78.9	71.6	117.2	78.0	40.0	86.1	38.5	83.3	37.2	80.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.0	(b) 16.8	(c) 14.9	(e) -3.0	(f) 95.5	(g) 5.6	(h) 3.4	(i) -7.0	(j) 97.9	(k) -10.3	(l) 99.9	(m) -13.4	(n) 101.8	(o) -17.4	(p) 104.2		
(a) 19.0	(b) 16.8	(c) 14.9	(e) -3.9	(f) 79.2	(g) 5.3	(h) 1.5	(i) -7.7	(j) 80.3	(k) -10.8	(l) 81.2	(m) -13.8	(n) 82.1	(o) -17.6	(p) 83.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	28.0	30.0	38.1	49.5	59.8	67.9	74.1	72.3	65.5	53.5	42.9	33.7
	DBStd	17.77	10.22	9.10	8.88	8.27	7.25	6.17	4.82	5.12	6.44	8.24	8.09	8.78
	HDD50	2546	684	562	382	105	5	0	0	0	58	241	509	
	HDD65	5748	1148	980	835	468	198	40	2	4	74	366	664	969
	CDD50	3054	1	1	12	92	308	537	747	692	464	167	28	5
	CDD65	781	0	0	0	5	36	127	284	231	88	10	0	0
	CDH74	6854	0	0	1	88	409	1131	2552	1944	667	61	1	0
	CDH80	2216	0	0	0	29	114	347	935	621	162	8	0	0
Wind	WSAvg	5.5	6.1	6.3	7.1	6.9	5.6	4.8	4.3	4.0	4.4	5.0	5.5	5.6
Precipitation	PrecAvg	52.6	3.9	3.2	4.7	4.2	4.0	4.3	4.4	4.6	5.1	5.0	4.5	4.5
	PrecMax	65.2	7.4	9.7	8.7	9.1	8.3	10.0	8.8	9.1	13.8	16.3	7.4	8.0
	PrecMin	39.8	1.9	1.6	0.7	0.9	0.5	0.7	1.8	1.5	1.6	0.6	1.4	1.4
	PrecStd	6.3	1.7	1.6	2.0	2.0	1.9	2.6	1.8	2.4	3.1	3.7	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	60.6	69.6	84.4	88.2	91.3	94.5	93.0	90.0	80.4	69.1	62.3
		MCWB	52.3	52.6	55.4	64.9	69.1	74.8	75.3	76.2	73.5	67.8	59.4	57.0
	2%	DB	52.3	53.6	62.5	74.7	83.1	87.6	91.0	89.4	84.3	74.8	64.2	56.4
		MCWB	47.9	47.6	50.8	57.6	66.8	71.9	73.9	73.3	69.6	65.3	57.4	52.8
	5%	DB	46.3	47.9	56.4	69.0	78.7	83.9	88.3	86.3	81.0	71.4	61.1	52.4
		MCWB	41.8	41.6	47.3	54.6	63.6	69.1	72.9	71.9	68.1	64.1	55.3	48.4
	10%	DB	42.4	43.2	52.0	64.0	73.8	80.9	85.3	83.4	77.7	67.5	57.2	47.9
		MCWB	37.8	37.6	44.9	52.1	61.4	67.8	71.4	70.5	66.8	61.0	51.7	43.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.9	54.8	58.7	65.7	72.0	76.5	78.9	78.9	75.6	71.5	63.3	58.0
		MCDB	56.1	57.2	65.2	83.5	84.7	88.0	89.9	90.0	84.3	76.3	66.2	60.1
	2%	WB	48.0	48.3	53.5	61.7	69.0	74.4	76.8	76.4	73.3	68.3	59.6	53.4
		MCDB	50.8	52.4	58.3	70.1	78.5	84.1	86.8	84.6	79.6	72.4	62.8	55.6
	5%	WB	42.8	42.2	49.1	58.1	66.7	72.4	75.2	74.9	71.6	65.2	56.7	49.2
		MCDB	46.1	46.5	55.5	64.5	75.3	80.3	83.5	82.0	77.2	69.8	60.0	52.6
	10%	WB	38.1	38.2	45.6	54.4	64.1	70.3	73.8	73.3	69.8	62.2	52.7	43.7
		MCDB	41.6	42.4	51.0	61.0	70.8	76.7	81.1	79.4	74.9	66.5	56.4	46.7
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.9	18.4	20.6	21.0	20.0	19.9	19.8	20.7	20.1	18.9	15.9
		MCDBR	20.9	23.7	26.0	28.3	28.2	24.8	23.3	22.3	23.7	23.2	23.3	20.4
		MCWBR	17.3	17.5	16.2	14.1	13.4	11.2	9.1	9.2	11.4	13.7	16.8	17.6
	5% WB	MCDBR	20.1	21.7	22.3	23.4	23.2	21.2	19.0	18.3	18.0	18.7	19.9	19.4
		MCWBR	18.0	18.0	16.3	13.4	12.6	10.8	8.7	8.7	10.7	12.9	17.0	18.4
Clear-Sky Solar Irradiance	taub	0.305	0.316	0.340	0.373	0.420	0.465	0.484	0.455	0.383	0.346	0.317	0.303	
	taud	2.455	2.433	2.430	2.368	2.263	2.175	2.133	2.206	2.380	2.498	2.537	2.531	
	Ebn at Noon	269	284	289	285	272	259	253	257	270	269	264	261	
	Edh at Noon	24	29	33	37	43	46	48	43	34	27	23	21	
All-Sky Solar Radiation	RadAvg	591	863	1173	1517	1694	1842	1887	1672	1379	924	647	488	
	RadStd	47	77	90	163	168	180	105	97	113	90	52	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+104

Nomenclature: See separate page